HOUSE PRICE DATASET

Exploratory Data Analysis (EDA) Project Report

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Batch:October batch (B)

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Dataset Source:Kaggle

1. Objective

The objective of this project is to perform exploratory data analysis on a house-price dataset using Python. EDA helps understand the structure and quality of the data, identify unusual values, summarize important variables, and explore relationships that may be useful for price prediction.

Main questions for analysis

• How are house prices distributed?
• Are there unusual values in house price and square footage?
• How does square footage relate to house price?
• How do bedrooms, bathrooms, year built, lot size, garage size, and neighborhood quality vary?
• What data-quality issues should be checked before modelling

3. Dataset Overview

First 5 Rows and column of the dataset using df.head() and df.tail()

	Square_Footage	Num_Bedrooms	Num_Bathrooms	Year_Built	Lot_Size	Garage_Size	Neighborhood_Quality	House_Price
0	1360.0	2	1	1981	0.599637	0	5	2.623829e+05
1	4272.0	3	3	2016	4.753014	1	6	9.852609e+05
2	3592.0	1	2	2016	3.634823	0	9	7.779774e+05
3	966.0	1	2	1977	2.730567	1	8	2.296989e+05
4	4926.0	2	1	1993	4.699073	0	8	1.041741e+06

Last 5 rows

	Square_Footage	Num_Bedrooms	Num_Bathrooms	Year_Built	Lot_Size	Garage_Size	Neighborhood_Quality	House_Price
1005	2755.0	2	2	1979	4.386386	1	4	616308.962420
1006	3620.0	5	3	2011	4.892841	1	8	861356.977559
1007	1471.0	5	1	1982	1.201914	0	3	341253.220780
1008	1743.0	2	1	1962	2.708443	0	5	344363.273682
1009	2016.0	1	1	2021	2.607486	0	3	450594.567977

4. Data Quality and Cleaning

Thenotebookchecks duplicate records withdf.duplicated().sum(). This gives the number of rows that are exact duplicates across all columns. The result should be recorded after the notebook is run.

```
df.duplicated().sum()
```

```
np.int64(10)
```

```
df.drop_duplicates(inplace=True)
```

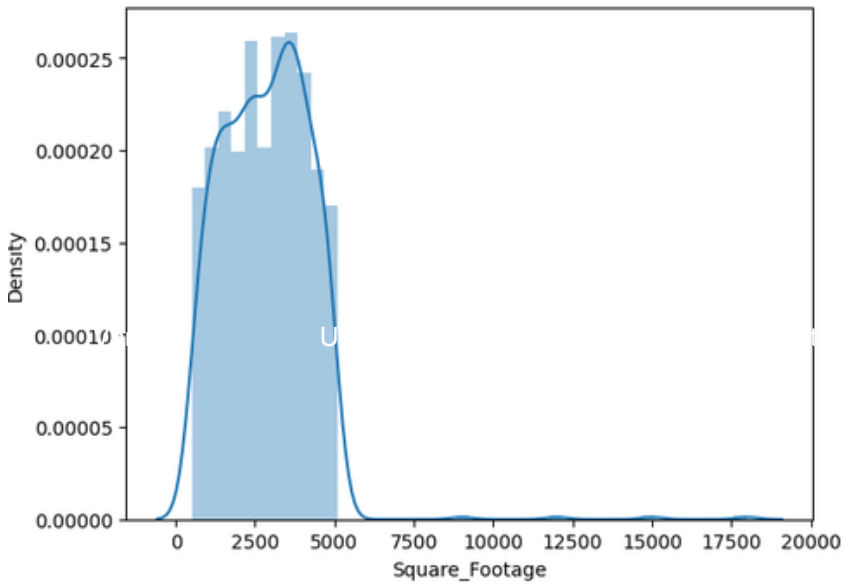
Outlier detection and treatment

Boxplots are created for price_usd and km_driven to inspect possible outliers. The notebook then applies the Interquartile Range (IQR) method to both columns:

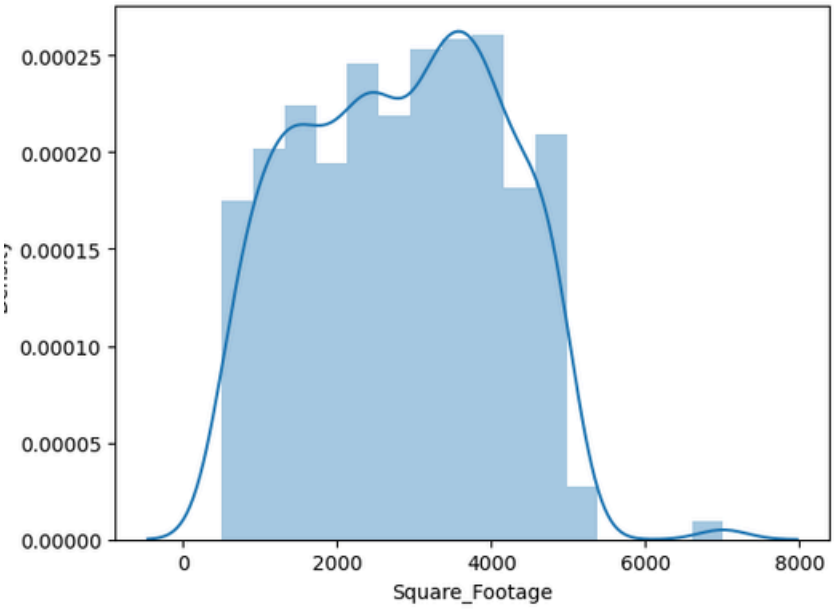
1. Calculate Q1 (25th percentile) and Q3 (75th percentile).
2. Calculate IQR = Q3 – Q1.
3. Calculate lower limit = Q1 – 1.5 × IQR and upper limit = Q3 + 1.5 × IQR.
4. Replace values outside these limits with the nearest limit (capping / winsorisation).

Duplicate rows	Output of df.duplicated().sum()
Missing values	Use df.isnull().sum() and record columns with missing data
Data types	Check df.info(); convert types only when justified
Outliers	Inspect boxplots for price_usd and km_driven
Treatment	IQR-based capping applied to price_usd and km_driven

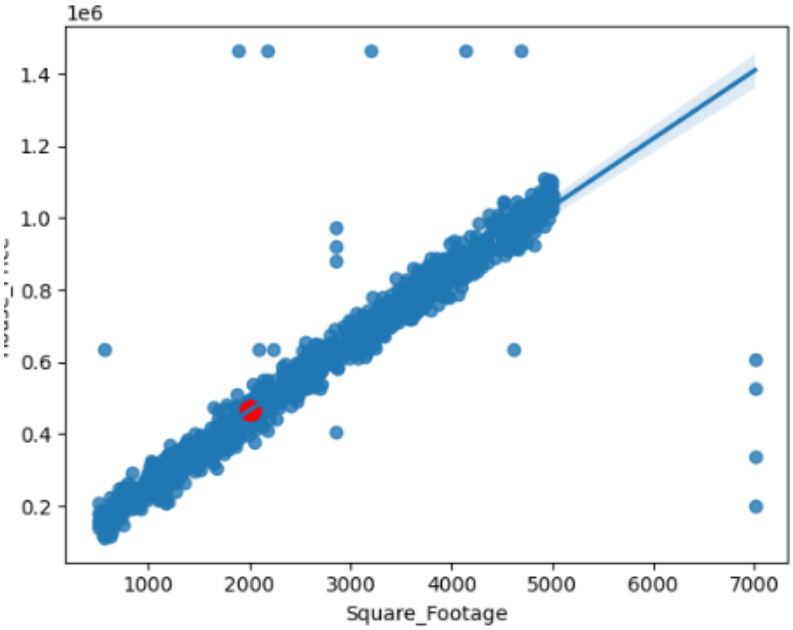
Before outlier treatment



After outlier treatment



Linear Regression Graph



5. Descriptive Analysis

Descriptive analysis summarizes the dataset before interpreting patterns. The following summary figures are based on the supplied CSV before removing duplicates or filling missing values. Re-run the notebook after cleaning to obtain final analysis values.

Selected numerical summary

Variable	Count	Mean	Median	Minimum	Maximum
Square_Footage	1,005	2,858.37	2,871	503	18,000
House_Price	1,005	633,519.97	629,609.10	111,626.85	5,000,000
Num_Bedrooms	1,010	2.99	3	1	5
Num_Bathrooms	1,010	1.97	2	1	3
Year_Built	1,010	1986.55	1986	1950	2022
Lot_Size	1,010	2.78	2.81	0.51	4.99
Garage_Size	1,010	1.02	1	0	2
Neighborhood_Quality	1,010	5.61	6	1	10

6. Visual EDA

Use the notebook's charts to explore the distributions of House_Price and Square_Footage, compare boxplots before and after IQR capping, and examine the relationship between property size and price with a regression plot.

Recommended charts

1. House price distribution — histogram / density plot; describe the shape and spread.
2. House price boxplot before cleaning — note potential outliers.
3. Square footage boxplot before cleaning — note potential outliers.
4. House price and square footage boxplots after IQR capping — compare with the original plots.
5. Square footage vs house price regression plot — describe the direction and spread of the relationship.
6. Add one more chart, such as a histogram of Year_Built or a bar chart of bedroom counts.

Important: report conclusions should be based on the charts generated by running the notebook.

Do not infer causation from visual relationships alone

7. Key Findings

The dataset contains 1,010 rows and 8 columns. Before cleaning, Square_Footage and House_Price each have 5 missing values, and 10 duplicate rows were identified. The price and square-footage fields contain high maximum values that should be reviewed using boxplots and the IQR method. Findings to discuss after running the notebook

1. Price distribution: compare the mean and median and describe whether the distribution is symmetric or skewed.
2. Price outliers: identify unusual values in the original House_Price boxplot.
3. Square-footage outliers: describe the spread and unusual property sizes.
4. Effect of capping: compare the before-and-after boxplots and note which extremes were capped.
5. Size and price: describe the direction and strength of the regression-plot pattern without assuming square footage alone determines price.

Limitations

• The dataset includes several property attributes, but the analysis may focus mainly on Square_Footage and House_Price for simple linear regression.
• Descriptive statistics shown here are calculated before cleaning; they may change after duplicate removal and missing-value treatment.
• Relationships in observational data do not prove causation. Price may also depend on bedrooms, bathrooms, year built, lot size, garage size, and neighborhood quality.
• IQR capping changes extreme values and may hide meaningful high- or low-priced properties.

8. Conclusion

This project demonstrates an EDA workflow for house-price data: load the CSV, inspect its structure, check duplicates and missing values, summarize numerical variables, visualise distributions, detect outliers using boxplots and the IQR rule, and explore the relationship between square footage and house price. Linear regression can then be trained with Square_Footage as the input feature and House_Price as the target.

Future improvements

Include a missing-value summary, descriptive-statistics table, labelled charts, and a comparison of findings before and after cleaning. For a more informative model, evaluate additional features such as bedrooms, bathrooms, year built, lot size, garage size, and neighborhood quality